

Sketching Parabolas from Standard Form

$$\text{Parabolas of the form } y = ax^2 + bx + c$$

Goal: Sketch the parabola and clearly show the *general shape* (happy/sad), the *y-intercept*, the *x-intercepts*, and the *turning point*.

Step 1: Identify a , b , and c .

Write the equation in the form $y = ax^2 + bx + c$ and read off:

$$a = \text{---}, \quad b = \text{---}, \quad c = \text{---}.$$

Step 2: General shape (happy or sad).

- If $a > 0$, the parabola opens upwards. You can describe it as **happy** (concave up).
- If $a < 0$, the parabola opens downwards. You can describe it as **sad** (concave down).

Step 3: y -intercept.

The y -intercept is where the graph crosses the y -axis. Here, $x = 0$.

$$y = a(0)^2 + b(0) + c = c.$$

So the y -intercept is the point $(0, c)$.

Step 4: x -intercepts.

The x -intercepts are where the graph crosses the x -axis. Here, $y = 0$.

$$0 = ax^2 + bx + c.$$

- Try **factorising** if possible.
- If it will not factorise nicely, use the **quadratic formula**:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

You may get:

- Two x -intercepts (two real solutions),
- One x -intercept (a repeated root, the parabola just touches the x -axis), or
- No real x -intercepts (the parabola does not cross the x -axis).

Step 5: Turning point.

The turning point is the highest or lowest point on the parabola.

- The x -coordinate of the turning point is

$$x_{\text{TP}} = -\frac{b}{2a}.$$

- Substitute this into the equation to find the y -coordinate:

$$y_{\text{TP}} = a \left(-\frac{b}{2a} \right)^2 + b \left(-\frac{b}{2a} \right) + c.$$

So the turning point is

$$(x_{\text{TP}}, y_{\text{TP}}) = \left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right) \right).$$

Step 6: Sketching the parabola.

- Plot the y -intercept.
- Plot any x -intercepts you found.
- Plot the turning point.
- Use the general shape (happy/sad) and symmetry about the vertical line $x = x_{\text{TP}}$ to sketch a smooth curve.

Worked example A (happy parabola).

Sketch $y = x^2 - 4x + 3$.

1. Identify a , b , c :

$$a = 1, \quad b = -4, \quad c = 3.$$

2. Shape:

$$a = 1 > 0 \Rightarrow \text{parabola is } \mathbf{happy} \text{ (opens up)}.$$

3. y -intercept:

$$y = c = 3 \Rightarrow (0, 3).$$

4. x -intercepts: solve $x^2 - 4x + 3 = 0$.

$$x^2 - 4x + 3 = (x - 1)(x - 3) = 0 \Rightarrow x = 1 \text{ or } x = 3.$$

So the x -intercepts are $(1, 0)$ and $(3, 0)$.

5. Turning point:

$$x_{\text{TP}} = -\frac{b}{2a} = -\frac{-4}{2 \cdot 1} = 2.$$

Substitute $x = 2$:

$$y_{\text{TP}} = 2^2 - 4 \cdot 2 + 3 = 4 - 8 + 3 = -1.$$

Turning point is $(2, -1)$.

6. Sketch using the points $(0, 3)$, $(1, 0)$, $(2, -1)$, $(3, 0)$ and a happy shape.

Worked example B (sad parabola).

Sketch $y = -x^2 - 2x + 3$.

1. Identify a , b , c :

$$a = -1, \quad b = -2, \quad c = 3.$$

2. Shape:

$$a = -1 < 0 \Rightarrow \text{parabola is } \mathbf{sad} \text{ (opens down)}.$$

3. y -intercept:

$$y = c = 3 \Rightarrow (0, 3).$$

4. x -intercepts: solve $-x^2 - 2x + 3 = 0$.

$$-x^2 - 2x + 3 = 0 \Rightarrow x^2 + 2x - 3 = 0 \Rightarrow (x + 3)(x - 1) = 0.$$

So $x = -3$ or $x = 1$, giving $(-3, 0)$ and $(1, 0)$.

5. Turning point:

$$x_{\text{TP}} = -\frac{b}{2a} = -\frac{-2}{2 \cdot (-1)} = \frac{2}{-2} = -1.$$

Substitute $x = -1$:

$$y_{\text{TP}} = -(-1)^2 - 2(-1) + 3 = -1 + 2 + 3 = 4.$$

Turning point is $(-1, 4)$.

6. Sketch using $(0, 3)$, $(-3, 0)$, $(1, 0)$, $(-1, 4)$ and a sad shape.

Sketching Parabolas from Turning-Point Form

Parabolas of the form $y = a(x - h)^2 + k$

Goal: Sketch the parabola and clearly show the *general shape* (happy/sad), the *turning point*, the *y-intercept*, and the *x-intercepts*.

Step 1: Identify a , h , and k .

Write the equation in the form $y = a(x - h)^2 + k$ and read off:

$$a = \text{----}, \quad h = \text{----}, \quad k = \text{----}.$$

Step 2: General shape (happy or sad).

- If $a > 0$, the parabola opens upwards (**happy**).
- If $a < 0$, the parabola opens downwards (**sad**).

Step 3: Turning point.

From $y = a(x - h)^2 + k$:

- The turning point is directly given as (h, k) .
- The axis of symmetry is the vertical line $x = h$.

Step 4: y-intercept.

Set $x = 0$ and substitute into the equation:

$$y = a(0 - h)^2 + k.$$

This gives the y -intercept at the point $(0, y)$.

Step 5: x-intercepts.

Set $y = 0$ and solve

$$0 = a(x - h)^2 + k.$$

- Rearrange to isolate $(x - h)^2$.
- Take square roots to find $x - h$.
- Solve for x to get one or two x -intercepts (or none, if there are no real solutions).

Step 6: Sketching the parabola.

- Plot the turning point (h, k) .
- Plot the y -intercept.
- Plot any x -intercepts found.
- Use the general shape (happy/sad) and the symmetry about $x = h$ to draw a smooth curve.

Worked example C (happy parabola).

Sketch $y = 2(x - 1)^2 - 3$.

1. Identify a , h , k :

$$a = 2, \quad h = 1, \quad k = -3.$$

2. Shape:

$$a = 2 > 0 \Rightarrow \text{parabola is } \mathbf{happy} \text{ (opens up)}.$$

3. Turning point:

$$(h, k) = (1, -3).$$

4. y -intercept: set $x = 0$.

$$y = 2(0 - 1)^2 - 3 = 2(1) - 3 = -1.$$

So the y -intercept is $(0, -1)$.

5. x -intercepts: set $y = 0$.

$$0 = 2(x - 1)^2 - 3 \Rightarrow 2(x - 1)^2 = 3 \Rightarrow (x - 1)^2 = \frac{3}{2}.$$

Take square roots:

$$x - 1 = \pm \sqrt{\frac{3}{2}} \Rightarrow x = 1 \pm \sqrt{\frac{3}{2}}.$$

(You can leave these as exact surds.)

6. Sketch using the turning point $(1, -3)$, the y -intercept $(0, -1)$ and symmetric x -intercepts.

Worked example D (sad parabola).

Sketch $y = -\frac{1}{2}(x + 2)^2 + 4$.

1. Identify a , h , k :

$$a = -\frac{1}{2}, \quad h = -2, \quad k = 4.$$

(Because $x + 2 = x - (-2)$.)

2. Shape:

$$a = -\frac{1}{2} < 0 \Rightarrow \text{parabola is } \mathbf{sad} \text{ (opens down)}.$$

3. Turning point:

$$(h, k) = (-2, 4).$$

4. y -intercept: set $x = 0$.

$$y = -\frac{1}{2}(0 + 2)^2 + 4 = -\frac{1}{2} \cdot 4 + 4 = -2 + 4 = 2.$$

So the y -intercept is $(0, 2)$.

5. x -intercepts: set $y = 0$.

$$0 = -\frac{1}{2}(x + 2)^2 + 4 \Rightarrow -\frac{1}{2}(x + 2)^2 = -4 \Rightarrow (x + 2)^2 = 8.$$

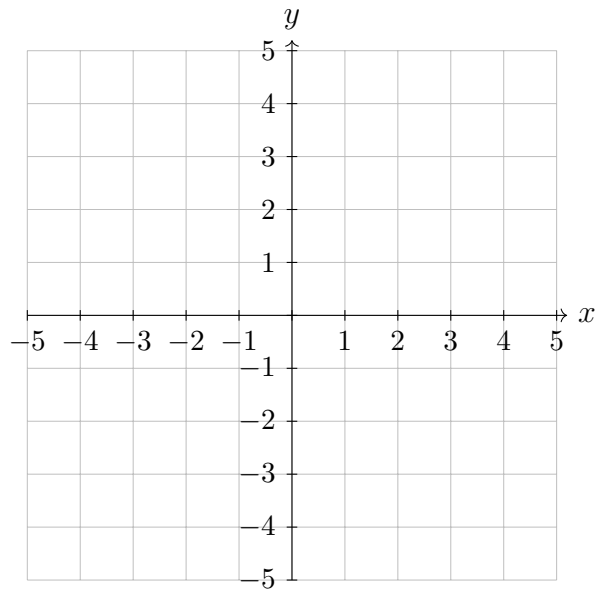
Take square roots:

$$x + 2 = \pm\sqrt{8} = \pm 2\sqrt{2} \Rightarrow x = -2 \pm 2\sqrt{2}.$$

6. Sketch using turning point $(-2, 4)$, y -intercept $(0, 2)$ and the two x -intercepts $(-2 \pm 2\sqrt{2}, 0)$.

Parabola Sketching Template

Standard Form: $y = ax^2 + bx + c$



Equation: _____

Identify the coefficients

$a =$ _____ $b =$ _____ $c =$ _____

1. General shape

Circle one: **happy** / **sad**

Reason: a is _____ (positive/negative).

2. y -intercept

Substitute $x = 0$ into $y = ax^2 + bx + c$:

$y = a(0)^2 + b(0) + c =$ _____

Point: $(0, \text{_____})$

3. x -intercepts

Solve $ax^2 + bx + c = 0$

(using factorising or the quadratic formula).

$x_1 =$ _____

$x_2 =$ _____

4. Turning point

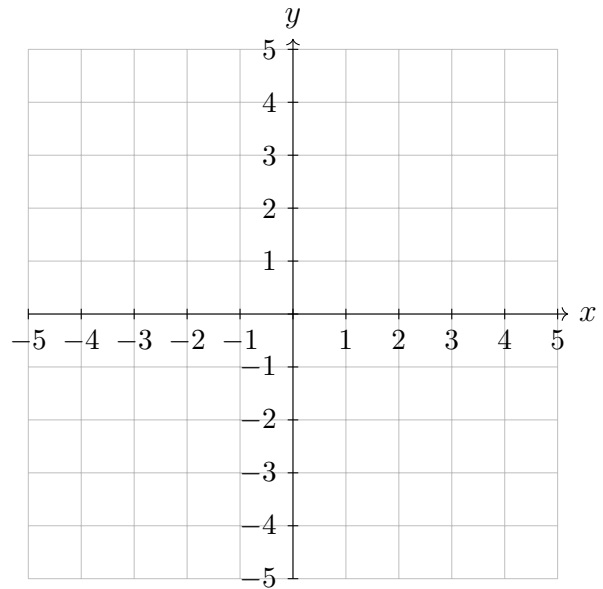
Use $x = -\frac{b}{2a}$:

$x_{\text{TP}} = -\frac{b}{2a} =$ _____

Substitute this x into the equation to find y :

$y_{\text{TP}} =$ _____

Turning point: $(\text{_____}, \text{_____})$



Equation: _____

Identify the coefficients

$a =$ _____ $b =$ _____ $c =$ _____

1. General shape

Circle one: **happy** / **sad**

Reason: a is _____ (positive/negative).

2. y -intercept

Substitute $x = 0$ into $y = ax^2 + bx + c$:

$y = a(0)^2 + b(0) + c =$ _____

Point: $(0, \text{_____})$

3. x -intercepts

Solve $ax^2 + bx + c = 0$

(using factorising or the quadratic formula).

$x_1 =$ _____

$x_2 =$ _____

4. Turning point

Use $x = -\frac{b}{2a}$:

$x_{\text{TP}} = -\frac{b}{2a} =$ _____

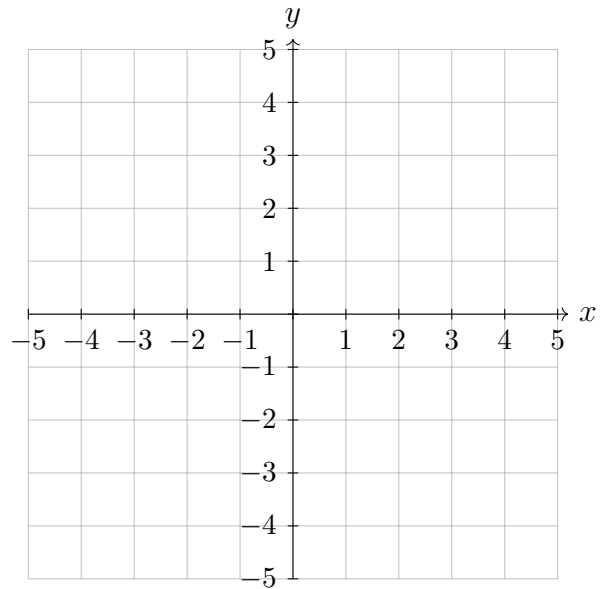
Substitute this x into the equation to find y :

$y_{\text{TP}} =$ _____

Turning point: $(\text{_____, } \text{_____})$

Parabola Sketching Template

Turning-Point Form: $y = a(x - h)^2 + k$



Equation: _____

Identify the parameters

$a =$ _____ $h =$ _____ $k =$ _____

1. General shape

Circle one: **happy** / **sad**

Reason: a is _____ (positive/negative).

2. Turning point

From $y = a(x - h)^2 + k$, the turning point is:

$(h, k) = (\text{_____}, \text{_____})$

3. y -intercept

Substitute $x = 0$:

$y = a(0 - h)^2 + k = \text{_____}$

Point: $(0, \text{_____})$

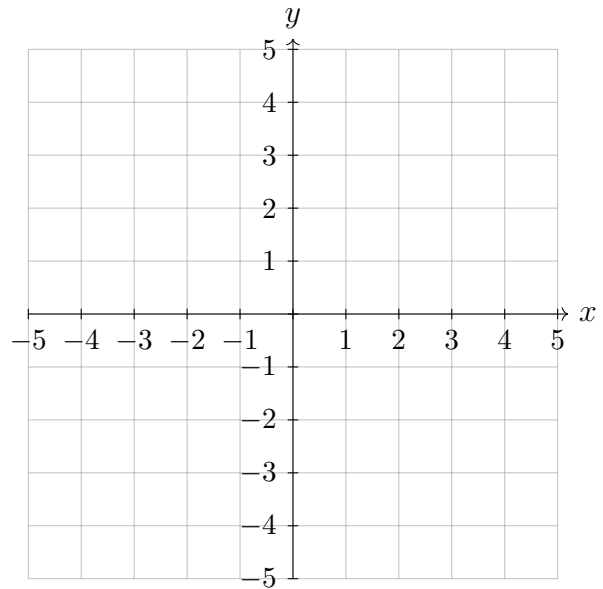
4. x -intercepts

Solve $a(x - h)^2 + k = 0$

$(x - h)^2 = \text{_____}$

$x_1 = \text{_____}$

$x_2 = \text{_____}$



Equation: _____

Identify the parameters

$a =$ _____ $h =$ _____ $k =$ _____

1. General shape

Circle one: **happy** / **sad**

Reason: a is _____ (positive/negative).

2. Turning point

From $y = a(x - h)^2 + k$, the turning point is:

$(h, k) = (\text{_____}, \text{_____})$

3. y -intercept

Substitute $x = 0$:

$y = a(0 - h)^2 + k = \text{_____}$

Point: $(0, \text{_____})$

4. x -intercepts

Solve $a(x - h)^2 + k = 0$

$(x - h)^2 = \text{_____}$

$x_1 = \text{_____}$

$x_2 = \text{_____}$